

24 Force acting on a single wire, $F = BIL = (0.010)(5.0 \times 10^{-3})(8 \times 10^{-3}) = 4 \times 10^{-7} \text{ N}$

Since there are 20 turns, force on each side of the loop, $F_{Total} = 20(4 \times 10^{-7}) = 8 \times 10^{-6} \text{ N}$

Torque of a couple = $F_{Total} (12.0 \times 10^{-3}) = 9.6 \times 10^{-8} \text{ N m}$

Answer: D

25 When the bar magnet is falling, the rate of decrease of magnetic flux linkage when leaving